
# A Case Study of the Beer Distribution System

Benjamin Hsiao-Frates,
George Rao, Lakhvir Sohal

# Problem Overview

- Demand for beer is projected to grow beyond Efes' current brewing capacity in the near future, so the company must plan for new facilities and/or expansions.
- Competitive pressure is increasing after Turk Tuborg's acquisition by Carlsberg, making it important for Efes to defend its dominant market position.
- Distribution expenses make up a large portion of total costs, so optimizing malt and beer transportation could generate significant savings.
- Efes wants to combine short-term distribution improvements with long-term strategic decisions about where to open and expand breweries.

# Dist Plan Improvement

| | |
|---|---|
| *Total shipping cost* | 11.8006524731429 |

**Min Z = $\sum i \sum j$ malt cost (i,j) * malt shipped (i,j) + $\sum j \sum k$ beer cost (i,j) * beer shipped (j,k)**

Where: i = malt plant, j = brewery, k = dist center

The total optimized shipping cost is now $11.8M, versus before it was $14.16M

**Total savings is now $2.36M, or 16.7% less cost**

**Changes:**

- Malt from Ayfton before was shipped almost entirely to Ankara and Import to Istanbul. The new plan now has Ayfton shipping to Istanbul, and Konya shipping to ankara
- With this, we eliminated all import malt since its shipping costs is more
- Beer distribution also changed to match distributors to closer breweries geographically

# Is it cost-effective for Efes to import malt? If not, when would it?


**Is it cost-effective to import malt? No:**

- Domestic plants (Afyon + Konya) have 98k tons of capacity, far exceeding the 55.4k tons needed
- Konya alone has 42.6k tons of unused capacity
- Import shipping costs are higher than domestic on every route

**When would importing become viable?**

- Import → Istanbul: shipping cost must drop by ~$0.00288M/1000 tons (~9% reduction)
- Import → Ankara: shipping cost must drop by ~$0.01426M/1000 tons (~44% reduction)
- Istanbul is the most likely entry point for imports if costs decrease

# Discuss the effects of beer demand variation on transportation costs using the information in the sensitivity report.

**Most expensive to serve:** Export/Izmir (shadow price $0.045M/ML): additional 1ML of export demand adds $45K in cost

**Cheapest to serve:** Istanbul (shadow price $0.003M/ML): only $3K per additional 1ML, since the Istanbul brewery ships locally at zero transport cost

**Allowable increase is 8ML across all DCs**: this is tied to Istanbul brewery's 8ML of slack capacity (250 - 242). Beyond this, the current routing structure breaks

**Allowable decrease varies significantly**: Antalya, Kayseri, Samsun, and Export can only drop 5ML before routes change, while Istanbul and Bursa have much wider flexibility

**Implication:** The solution is most sensitive to demand drops at smaller southern/eastern DCs, and overall system capacity limits growth to just 8ML before expansion is needed

# Optimal Capacity Expansion Plan: New Brewery Locations & Investment

| Objective (Minimize total cost, million $) | |
|---|---|
| Malt transport cost | 1.7407 |
| Beer transport cost | 9.8286 |
| Fixed investment cost | 12.0000 |
| TOTAL COST | 23.5692 |

**Total cost: $23.56M**

**Open 3 new breweries:**

- **Izmir**: Open + Expand (120ML capacity, $6.0M/yr fixed cost)
- **Samsun**: Open only (90ML capacity, $2.5M/yr)
- **AnkNew**: Open only (90ML capacity, $3.5M/yr)

**Why these locations?**

- Izmir: serves Izmir demand and exports at near-zero transport cost; cheapest malt import route
- Samsun: serves Samsun demand at zero transport cost, as well as having the lowest fixed cost ($2.5M)
- AnkNew: provides remaining needed at low fixed cost ($3.5M), and serves Kayseri and Export overflow

# Non-Optimal Scenarios

| Change | Total Cost | Difference from Optimal |
|---|---|---|
| **Remove AnkNew** | 23.62M | +0.05M |
| **Remove Samsun** | 27.05M | +3.48M |
| **Don’t expand Izmir** | 23.86M | +0.29M |
| **Don’t open Samsun or AnkNew** | 27.18M | +3.61M |

**Changes:**

- **Remove AnkNew:** Sakarya substitutes almost perfectly, these two are interchangeable, so other external factors could determine the choice
- **Remove Samsun:** Sakarya opens but can't serve Samsun demand locally, forcing expensive long-distance shipping
- **Don't expand Izmir:** Three other breweries open to compensate, but the $1M expansion is far cheaper than building new capacity elsewhere
- **Don't open Samsun or AnkNew:** Forces Adana and Sakarya open, but Adana's remote southeast location drives up transport costs significantly

**Takeaway:** Izmir and Samsun are non-negotiable, they drive the most savings. AnkNew is the most flexible decision and could be swapped for Sakarya at nearly zero cost.

# Opening breweries below full capacity

| Capacity % | Total Cost | Change |
|---|---|---|
| 95 | 23.81M | +.24M |
| 90 | 24.47M | + .90M |
| 85 | 25.27M | +1.70M |
| 80 | 27.16M | +3.59M |

**Takeaways if forced to open below capacity:**

- **95%** utilization is recommended: only +$0.24M for a 5% buffer against demand spikes, maintenance, and quality issues
- Costs increase gradually to 90%, then accelerate sharply as the model needs to open additional breweries
- Below 85% is not cost-justified: fixed costs balloon as 4+ breweries are required to meet demand at reduced capacity
- **Recommendation:** Operate at 95%: minimal cost for meaningful operational flexibility